

Controlling for sugar and ascorbic acid, a mixture of flavonoids matching navel oranges significantly increases human postprandial serum antioxidant...

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Fruit and vegetable consumption reduces the risk for cardiovascular disease development. The postprandial state is an important contributor to chronic disease development. Orange flavonoids may reduce postprandial oxidation. It was hypothesized that a mixture of orange flavonoids would reduce postprandial oxidation better than a single orange flavonoid or orange sugar and ascorbic acid, but not as well as orange juice, when consumed with a typical breakfast. A placebo-controlled crossover trial (16 male and female participants, 4 treatments, 4 visits) was carried out. Treatments were placebo (ascorbic acid and sugar equivalent to orange juice); placebo plus hesperidin; placebo plus hesperidin, luteolin, and naringenin (mixture; found to have synergistic antioxidant properties in vitro in previous work); and orange juice (positive control). Serum oxygen radical absorbance capacity (ORAC), total plasma phenolics (TP), and serum lipoprotein oxidation (LO) were measured after a 12-hour baseline fast and at 1, 2, and 3 hours after sample consumption. The placebo plus mixture and orange juice groups were significantly increased in ORAC and LO lag time. Data for TP were inconsistent with ORAC and LO. Contrary to previous studies attributing the protective postprandial effect to fructose and ascorbate in other fruit trials, orange phenolic compounds contribute directly to the postprandial oxidative protection of serum, despite an inconsistent change in serum TP. Copyright © 2011 Elsevier Inc. All rights reserved.